

Predicting Falcon 9 First-Stage Landing Success

SpaceX launch data · EDA · SQL · Folium · Dash · Machine Learning

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June 2026

GitHub Repository Link

PUBLIC GITHUB REPOSITORY - REQUIRED EVIDENCE

- Repository URL: <https://github.com/massiel19em/Applied-Data-Science-Capstone>
- Contains the completed notebooks and Python files used throughout the capstone.
- Methodology source files are clearly listed: API, web scraping, wrangling, EDA visualization, SQL, Folium, Dash app, and MLprediction.
- Use this same repository URL in the final PDF submission so the evaluator can open the evidence directly.

Recommended files: 1. API notebook · 2. Web scraping notebook · 3. Data wrangling notebook · 4. EDA dataviz · 5. EDA SQL · 6.Folium · spacex_dash_app.py · 7. ML prediction notebook

Report Outline

01 **Executive Summary**

02 **Introduction**

03 **Methodology**

04 **Results**

05 **Discussion**

06 **Conclusion**

A technical story from raw SpaceX launch data to model-driven landing predictions.

What this project proves

The objective is to predict whether the Falcon 9 first stage will land successfully using classification algorithms. The project moves through data collection, wrangling, exploratory analysis, interactive visualization, and machine learning. EDA suggests that launch features such as payload mass, orbit type, and launch site are related to mission outcome. Decision Tree achieved the strongest GridSearchCV score among the tested models, while test accuracy was similar across models.

Business question and prediction target

SpaceX can reduce launch cost because the Falcon 9 first stage may be reused after a successful landing. If the landing outcome can be predicted, an alternate provider can estimate launch economics and bid more competitively.

Some unsuccessful landings are planned controlled ocean landings, so the analysis focuses on the final mission outcome. The core question: for a given launch profile, will the first stage land successfully?

End-to-end data science workflow

Data collection and wrangling: SpaceX API, Wikipedia web scraping, missing value handling, and feature engineering.
Exploratory analysis: Pandas, NumPy, SQL queries, launch-site counts, orbit distributions, and mission outcomes.
Interactive visualization: Matplotlib, Seaborn, Folium maps, and Plotly Dash dashboard components.
Machine learning prediction: Logistic Regression, SVM, Decision Tree, and KNN with hyperparameter tuning.

Data collection from SpaceX API

SpaceX API

- SpaceX API

- The API used is <https://api.spacexdata.com/v4/rockets/>.
- The API provides data about many types of rocket launches done by SpaceX, the data is therefore filtered to include only Falcon 9 launches.
- Every missing value in the data is replaced the mean the column that the missing value belongs to.
- We end up with 90 rows or instances and 17 columns or features. The picture below shows the first few rows of the data:

FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial	Longitude	Latitude	
4	1	2010-06-04	Falcon 9	NaN	LEO	CCSFS SLC 40	None None	1	False	False	False	None	1.0	0	B0003	-80.577366	28.561857
5	2	2012-05-22	Falcon 9	525.0	LEO	CCSFS SLC 40	None None	1	False	False	False	None	1.0	0	B0005	-80.577366	28.561857
6	3	2013-03-01	Falcon 9	677.0	ISS	CCSFS SLC 40	None None	1	False	False	False	None	1.0	0	B0007	-80.577366	28.561857
7	4	2013-09-29	Falcon 9	500.0	PO	VAFB SLC 4E	False Ocean	1	False	False	False	None	1.0	0	B1003	-120.610829	34.632093
8	5	2013-12-03	Falcon 9	3170.0	GTO	CCSFS SLC 40	None None	1	False	False	False	None	1.0	0	B1004	-80.577366	28.561857

API collection process

- Retrieved launch-level data from the SpaceX API and filtered the records to Falcon 9 missions.
- Selected operational features such as payload mass, orbit, launch site, booster version, coordinates, and outcome.
- Output: a structured launch dataset used as the foundation for wrangling and EDA.

Source: 1. [jupyter-labs-spacex-data-collection-api.ipynb](#)

Data collection from web scraping

• Web scraping

- The data is scraped from [https://en.wikipedia.org/w/index.php?title=List of Falcon 9 and Falcon Heavy launches&oldid=1027686922](https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922)
- The website contains only the data about Falcon 9 launches.
- We end up with 121 rows or instances and 11 columns or features. The picture below shows the first few rows of the data:

	Flight No.	Launch site	Payload	Payload mass	Orbit	Customer	Launch outcome	Version Booster	Booster landing	Date	Time
0	1	CCAFS	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success\n	F9 v1.0B0003.1	Failure	4 June 2010	18:45
1	2	CCAFS	Dragon	0	LEO	NASA	Success	F9 v1.0B0004.1	Failure	8 December 2010	15:43
2	3	CCAFS	Dragon	525 kg	LEO	NASA	Success	F9 v1.0B0005.1	No attempt\n	22 May 2012	07:44
3	4	CCAFS	SpaceX CRS-1	4,700 kg	LEO	NASA	Success\n	F9 v1.0B0006.1	No attempt	8 October 2012	00:35
4	5	CCAFS	SpaceX CRS-2	4,877 kg	LEO	NASA	Success\n	F9 v1.0B0007.1	No attempt\n	1 March 2013	15:10

Web scraping process

- Scraped Falcon 9 launch tables from Wikipedia to complement the API dataset.
- Parsed HTML tables and standardized launch date, booster, payload, orbit, customer, and landing fields.
- Output: an additional historical dataset for validating launch records and enriching analysis.

Source: 2. [jupyter-labs-webscraping.ipynb](#)

Data wrangling and feature preparation

Missing entries were handled so that the modeling dataset could be used. Categorical variables were encoded using one-hot encoding to make them numeric. A binary target column called Class was created: 0 = failed launch, 1 = successful launch. The final feature table contains 90 instances and 83 features.

Wrangling output

- Missing payload values were handled so models could train on consistent numeric input.
- Categorical fields were converted through one-hot encoding before classification.
- The final matrix has 90 launch records and 83 model-ready features plus the binary Class target.

Source: 3. [labs-jupyter-spacex-Data wrangling.ipynb](#)

EDA with Pandas, NumPy, and SQL



- Pandas and NumPy
 - Functions from the Pandas and NumPy libraries are used to derive basic information about the data collected, which includes:
 - The number of launches on each launch site
 - The number of occurrence of each orbit
 - The number and occurrence of each mission outcome
- SQL
 - The data is queried using SQL to answer several questions about the data such as:
 - The names of the unique launch sites in the space mission
 - The total payload mass carried by boosters launched by NASA (CRS)
 - The average payload mass carried by booster version F9 v1.1



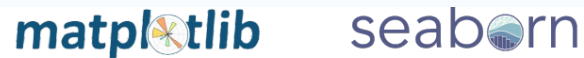
EDA questions answered

- How many launches occurred at each site?
- Which orbit and mission outcomes appear most frequently?
- SQL was used to validate launch-site, payload, booster, and landing-outcome summaries.

Sources: 4. EDA dataviz notebook · 5. EDA SQL notebook

Visualization with Matplotlib, Seaborn, and Folium

- Matplotlib and Seaborn



- Functions from the Matplotlib and Seaborn libraries are used to visualize the data through scatterplots, bar charts, and line charts.
- The plots and charts are used to understand more about the relationships between several features, such as:
 - The relationship between flight number and launch site
 - The relationship between payload mass and launch site
 - The relationship between success rate and orbit type

- Folium



- Functions from the Folium libraries are used to visualize the data through interactive maps.
- The Folium library is used to:
 - Mark all launch sites on a map
 - Mark the succeeded launches and failed launches for each site on the map
 - Mark the distances between a launch site to its proximities such as the nearest city, railway, or highway

Visualization methodology

- Matplotlib and Seaborn were used for scatterplots, bar charts, and yearly trend charts.
- Folium mapped launch sites, success/failure markers, and distance to nearby infrastructure.
- The goal was to connect technical features to visible outcome patterns before modeling.

Sources: 4. EDA dataviz notebook · 6. Folium notebook

Interactive analytics with Plotly Dash

Dashboard

03



- Dash
 - Functions from Dash are used to generate an interactive site where we can toggle the input using a dropdown menu and a range slider.
 - Using a pie chart and a scatterplot, the interactive site shows:
 - The total success launches from each launch site
 - The correlation between payload mass and mission outcome (success or failure) for each launch site

Interactive analytics

- Dash added user-driven filtering by launch site and payload mass range.
- The dashboard combines pie and scatter charts to explore success distribution interactively.
- This makes the project easier to inspect beyond static notebook outputs.

Source: `spacex_dash_app.py`

Machine learning prediction workflow

- Functions from the Scikit-learn library are used to create our machine learning models.
- The machine learning prediction phase include the following steps:
 - Standardizing the data
 - Splitting the data into training and test data
 - Creating machine learning models, which include:
 - Logistic regression
 - Support vector machine (SVM)
 - Decision tree
 - K nearest neighbors (KNN)
 - Fit the models on the training set
 - Find the best combination of hyperparameters for each model
 - Evaluate the models based on their accuracy scores and confusion matrix



Classification methodology

- Features were standardized and split into training and test sets.
- Four classifiers were tested: Logistic Regression, SVM, Decision Tree, and KNN.
- GridSearchCV tuned hyperparameters; models were evaluated with accuracy and confusion matrices.

Source: 7. *SpaceX_Machine_Learning_Prediction.ipynb*

How the results are organized

SQL results summarize launch sites, payload mass, booster versions, and landing outcomes.

Visualization results explain relationships between flight number, payload mass, launch site, orbit, and success rate.

Folium maps add geographic context for launch sites, outcomes, and distances to nearby infrastructure.

Dash results show interactive filtering for launch-site success rates and payload ranges.

Predictive analysis compares four classification models using accuracy and GridSearchCV scores.

Unique launch sites and CCA records

- The names of the unique launch sites in the space mission

Launch_Sites

CCAFS LC-40

CCAFS SLC-40

KSC LC-39A

VAFB SLC-4E

- 5 records where launch sites begin with 'CCA'

DATE	time__utc__	booster_version	launch_site	payload	payload_mass__kg__	orbit	customer	mission_outcome	landing__outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	07:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	00:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

SQL result

- Four unique launch sites were identified for the mission dataset.
- CCA-prefixed records confirm early launch activity at CCAFS LC-40.
- This validates launch_site as a key categorical input for EDA and modeling.

Payload mass and first ground-pad landing

SQL

- The total payload mass carried by boosters launched by NASA (CRS)

Total payload mass by NASA (CRS)
45596

- The average payload mass carried by booster version F9 v1.1

Average payload mass by Booster Version F9 v1.1
2928

- The date when the first successful landing outcome in ground pad was achieved

Date of first successful landing outcome in ground pad
2015-12-22

SQL result

- NASA CRS missions carried a total payload mass of 45,596 kg.
- F9 v1.1 average payload mass was 2,928 kg.
- First successful ground-pad landing was identified on 2015-12-22.

Drone-ship successes and mission outcomes

SQL

- The names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

booster_version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

- The total number of successful and failure mission outcomes

number_of_success_outcomes	number_of_failure_outcomes
100	1

SQL result

- Drone-ship successes with payloads between 4,000 and 6,000 kg were isolated by booster version.
- Mission outcome counts show successful missions strongly outnumber failures in this query.
- These checks support the success/failure framing for classification.

Booster versions with maximum payload mass

SQL

- The names of the booster versions which have carried the maximum payload mass

booster_version

F9 B5 B1048.4

F9 B5 B1048.5

F9 B5 B1049.4

F9 B5 B1049.5

F9 B5 B1049.7

F9 B5 B1051.3

F9 B5 B1051.4

F9 B5 B1051.6

F9 B5 B1056.4

F9 B5 B1058.3

F9 B5 B1060.2

F9 B5 B1060.3

SQL result

- Maximum-payload missions are concentrated in later Falcon 9 Block 5 booster versions.
- The query identifies booster versions associated with the heaviest missions.
- Booster version can therefore help explain mission capability and outcomes.

Failed drone-ship outcomes and landing counts

- The failed landing outcomes in drone ship, their booster versions, and launch site names for in year 2015

DATE	booster_version	launch_site
2015-01-10	F9 v1.1 B1012	CCAFS LC-40
2015-04-14	F9 v1.1 B1015	CCAFS LC-40

- The count of landing outcomes between the date 2010-06-04 and 2017-03-20, in descending order

landing__outcome	landing_count
No attempt	10
Failure (drone ship)	5
Success (drone ship)	5
Controlled (ocean)	3
Success (ground pad)	3
Failure (parachute)	2
Uncontrolled (ocean)	2
Precluded (drone ship)	1

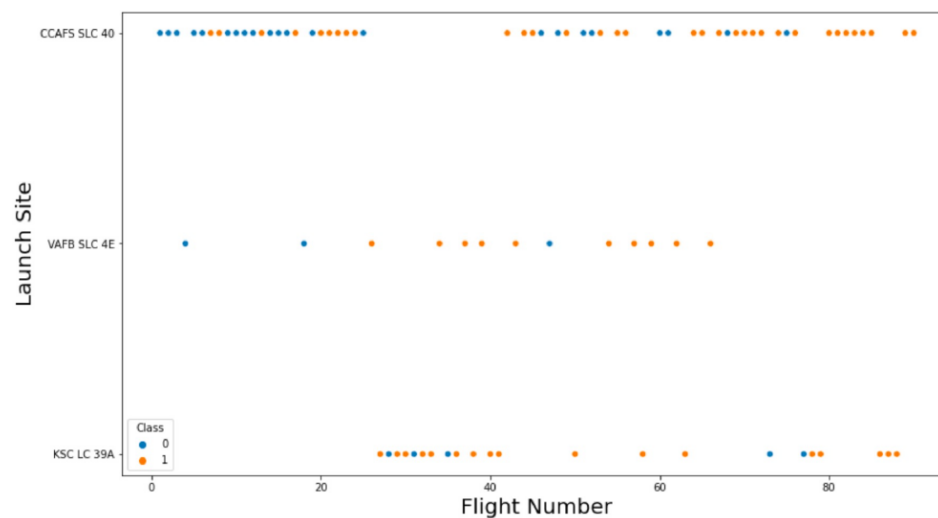
SQL result

- 2015 drone-ship failures occurred at CCAFS LC-40.
- Between 2010 and 2017, no-attempt landings were the most frequent outcome.
- Landing outcome categories show why the target variable needed careful definition.

Flight number vs. launch site

Viz

- The relationship between flight number and launch site



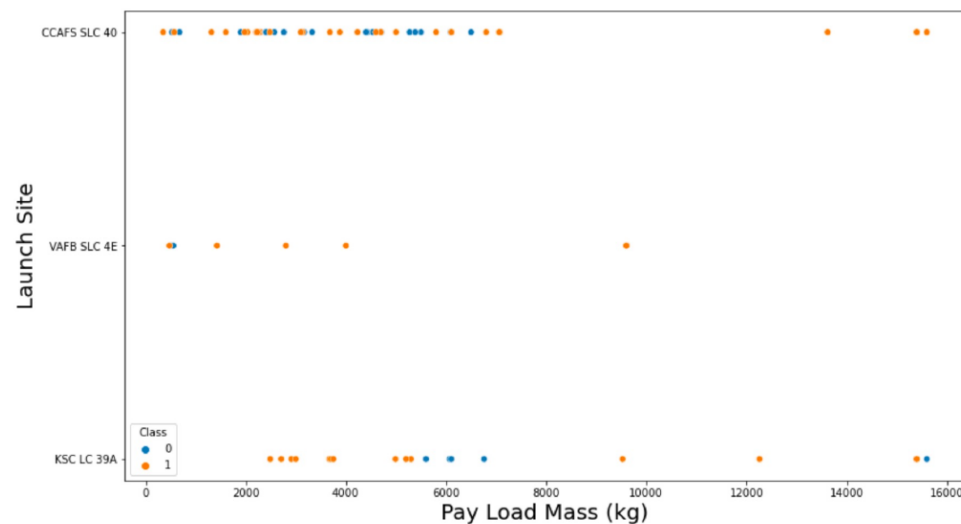
EDA insight

- Flight number captures the project learning curve over time.
- Later flights show more successful landing outcomes across launch sites.
- Launch site and flight number both contribute useful signal.

Payload mass vs. launch site

Viz

- The relationship between payload mass and launch site



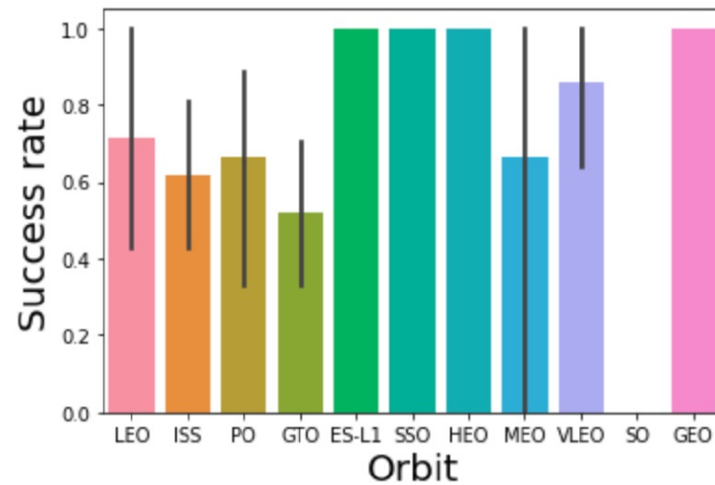
EDA insight

- Payload mass varies meaningfully by launch site.
- Some sites show success even at higher payload mass levels.
- The relationship is site-dependent rather than purely linear.

Success rate by orbit type

Viz

- The relationship between success rate and orbit type



EDA insight

- Orbit type displays different success-rate patterns.
- Some orbits are associated with higher landing success than others.
- Orbit is therefore an important explanatory feature for classification.

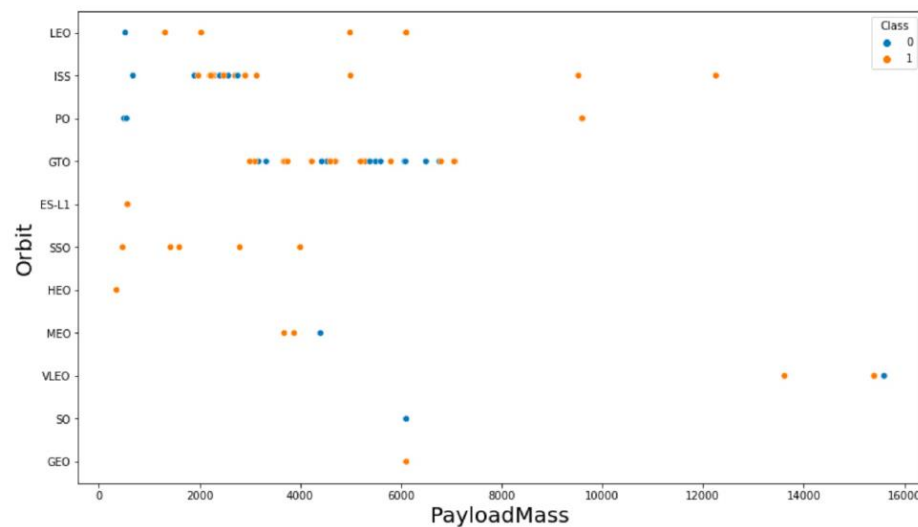
Flight number vs. orbit type

- The relationship between flight number and orbit type

Payload mass vs. orbit type

Viz

- The relationship between payload mass and orbit type



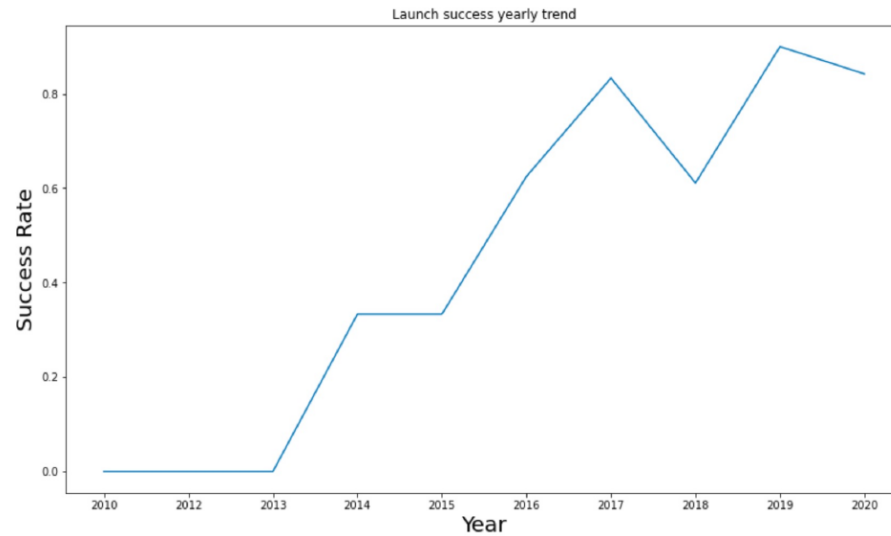
EDA insight

- Payload mass alone does not fully explain landing outcome.
- Its relationship with success changes depending on orbit type.
- This interaction motivated the use of supervised machine learning.

Launch success yearly trend

Viz

- The launch success yearly trend



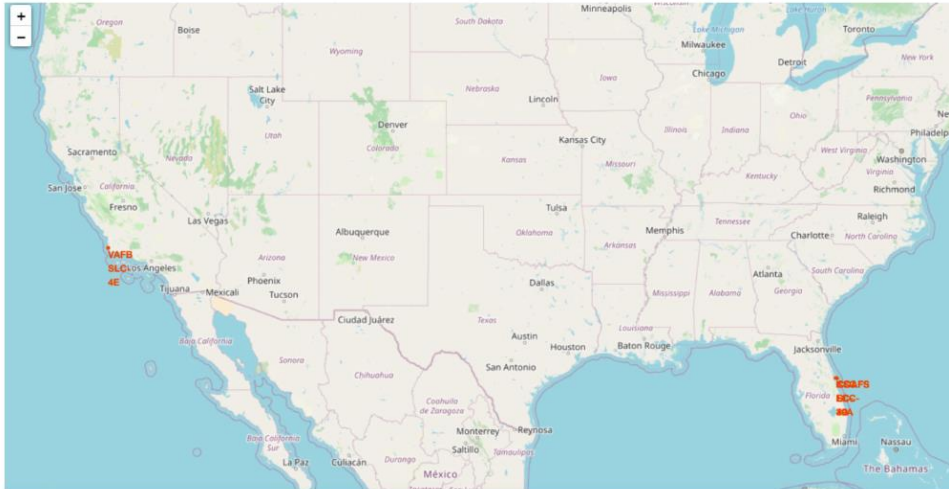
EDA insight

- The yearly trend suggests landing success improved over time.
- This is consistent with a learning-curve effect in reusable launch operations.
- Newer launch data should be prioritized for future validation.

All launch sites on map

Map

- All launch sites on map



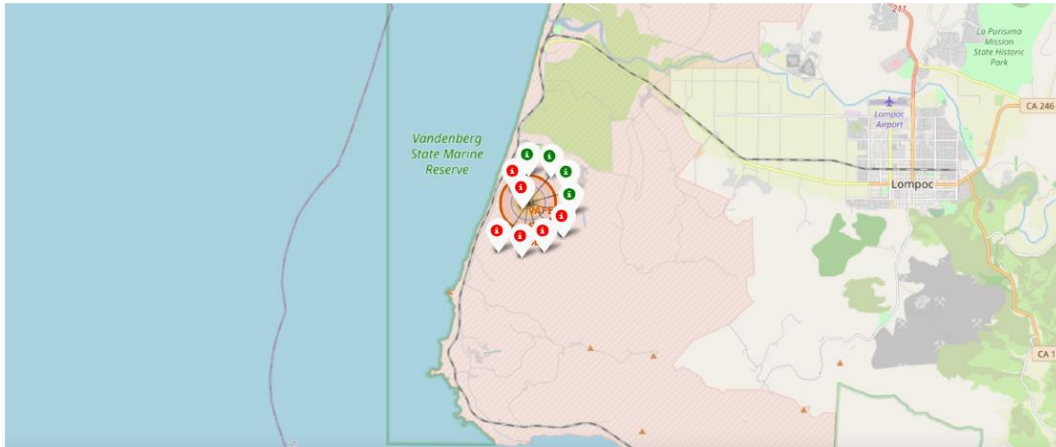
Folium result

- Launch sites are geographically concentrated along coastal areas.
- Mapping provides context that cannot be seen in tabular data alone.
- Location is linked to operational constraints and landing options.

Success and failure markers by site

Map

- The succeeded launches and failed launches for each site on map
 - If we zoom in on one of the launch site, we can see green and red tags. Each green tag represents a successful launch while each red tag represents a failed launch



Folium result

- Success and failure markers make site-level outcome patterns visible.
- The map shows where historical landing outcomes differ by location.
- This supports launch-site segmentation in the final model.

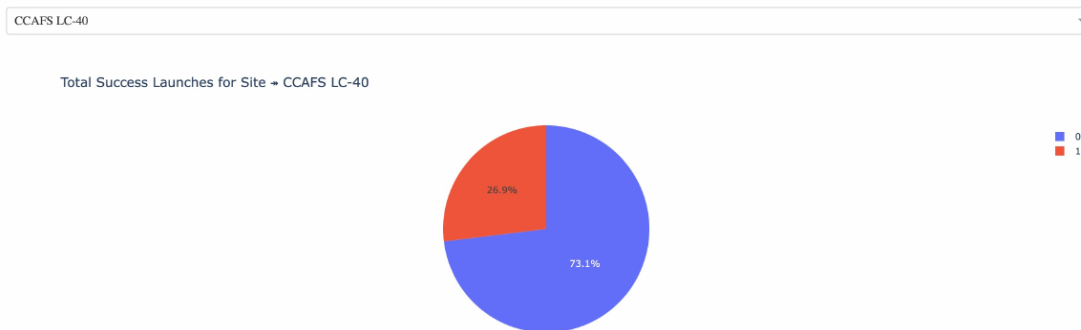
Launch-site success distribution

Dash

04

- The picture below shows a pie chart when launch site CCAFS LC-40 is chosen.
- 0 represents failed launches while 1 represents successful launches. We can see that 73.1% of launches done at CCAFS LC-40 are failed launches.

SpaceX Launch Records Dashboard



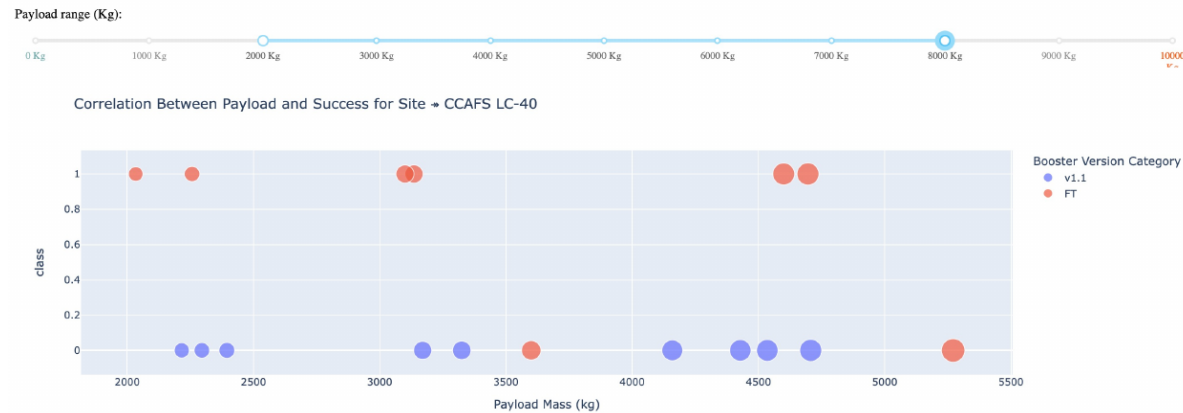
Dash result

- The dashboard summarizes total successful launches by launch site.
- Dropdown filtering allows site-level comparison instead of one static view.
- Interactive analytics improves how stakeholders explore the results.

Payload mass range and mission outcome

Dash

- The picture below shows a scatterplot when the payload mass range is set to be from 2000kg to 8000kg.
- Class 0 represents failed launches while class 1 represents successful launches.



Dash result

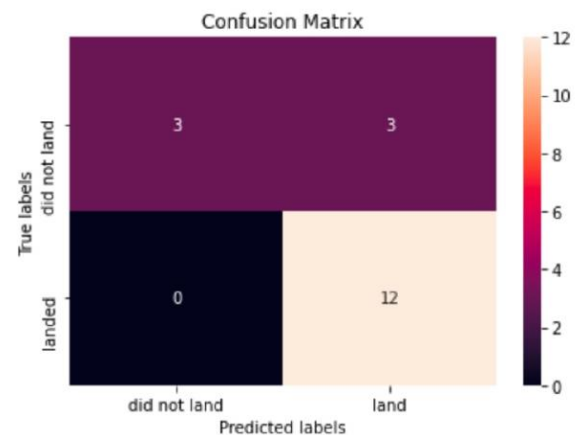
- Payload range filtering shows how mission outcome changes across payload mass.
- The scatterplot links payload, booster version, and launch site interactively.
- Dash turns EDA into a reusable analytical tool.

Logistic Regression results

ML

- Logistic regression

- GridSearchCV best score: 0.8464285714285713
- Accuracy score on test set: 0.8333333333333334
- Confusion matrix:



Model result

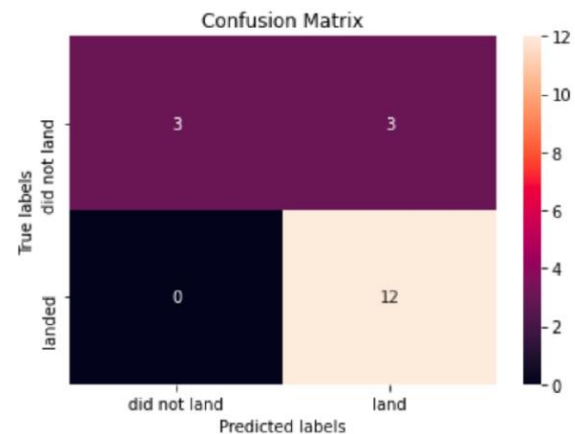
- Logistic Regression achieved test accuracy of 0.8333.
- Confusion matrix shows correct predictions for 12 landed cases and 3 non-landed cases.
- It provides an interpretable baseline for classification.

SVM results

ML

04

- Support vector machine (SVM)
 - GridSearchCV best score: 0.8482142857142856
 - Accuracy score on test set: 0.8333333333333334
 - Confusion matrix:



Model result

- SVM achieved test accuracy of 0.8333.
- The confusion matrix pattern was similar to Logistic Regression.
- Its GridSearchCV score was slightly stronger than Logistic Regression.

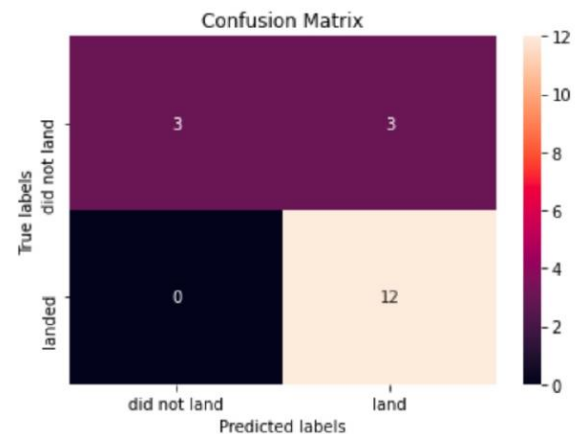
Decision Tree results

ML

04

- Decision tree

- GridSearchCV best score: 0.8892857142857142
- Accuracy score on test set: 0.8333333333333334
- Confusion matrix:



Model result

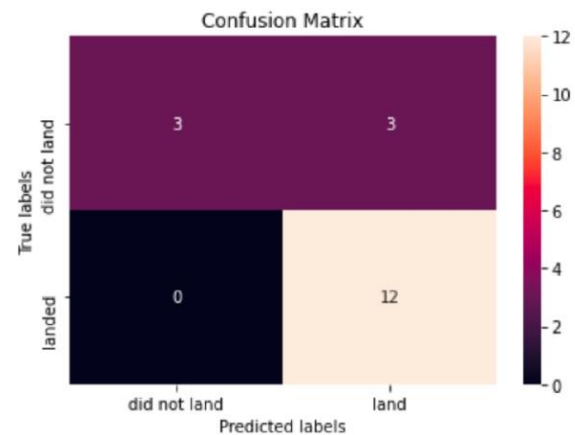
- Decision Tree achieved test accuracy of 0.8333.
- It ranked highest by GridSearchCV best score: 0.8893.
- This made it the selected best model for the final ranking.

KNN results

ML

04

- K nearest neighbors (KNN)
 - GridSearchCV best score: 0.8482142857142858
 - Accuracy score on test set: 0.8333333333333334
 - Confusion matrix:



Model result

- KNN achieved test accuracy of 0.8333.
- It ranked second by GridSearchCV best score: 0.8482.
- The result confirms several models can capture similar patterns.

Model ranking using GridSearchCV

ML

04

- Putting the results of all 4 models side by side, we can see that they all share the same accuracy score and confusion matrix when tested on the test set.
- Therefore, their GridSearchCV best scores are used to rank them instead. Based on the GridSearchCV best scores, the models are ranked in the following order with the first being the best and the last one being the worst:
 1. Decision tree (GridSearchCV best score: 0.8892857142857142)
 2. K nearest neighbors, KNN (GridSearchCV best score: 0.8482142857142858)
 3. Support vector machine, SVM (GridSearchCV best score: 0.8482142857142856)
 4. Logistic regression (GridSearchCV best score: 0.8464285714285713)

Model ranking

- All four models had the same test accuracy, so GridSearchCV scores were used for ranking.
- Final ranking: Decision Tree, KNN, SVM, Logistic Regression.
- Decision Tree is the strongest selected classifier in this experiment.

Interpreting the analytical evidence

Several variables appear to influence landing outcome, including orbit type, payload mass, and launch-site characteristics. For some orbit types, heavier payloads still show positive landing outcomes; for others, the pattern is less clear. Because the relationships are not always visually separable, supervised learning provides a structured way to learn patterns from historical data. The final model should be interpreted as a decision-support tool, not as a deterministic guarantee of launch outcome.

Final takeaway

This capstone predicts Falcon 9 first-stage landing outcome to support launch-cost estimation. The project applied the full data science pipeline: collection, wrangling, EDA, SQL, interactive analytics, and classification modeling. The models achieved similar test-set accuracy; Decision Tree ranked highest using GridSearchCV best score. A strong next step would be to validate the model with newer launches and enrich it with additional operational variables.

Data wrangling and feature preparation

1. Clean and standardize launch records

Filtered the raw API data to focus on Falcon 9 launches only.

Standardized launch-site, orbit, booster, payload, and outcome fields.

Handled missing payload values so the final dataset could be modeled consistently.

2. Define the prediction target

Created the binary Class variable from landing outcome.

Class = 1 identifies a successful first-stage landing.

Class = 0 identifies a failed or unsuccessful landing outcome.

3. Build the modeling matrix

Converted categorical features with one-hot encoding.

Separated X features from y target before training.

Final feature table: 90 launch records and 83 model-ready variables.

[GitHub source: 3. labs-jupyter-spacex-Data wrangling.ipynb](#)

EDA visualization and interactive analytics

Visualization questions

How does flight number relate to landing success by launch site?
How does payload mass vary across launch sites and orbit types?
Which orbit categories show higher or lower success rates over time?

Tools and outputs

Matplotlib and Seaborn: scatterplots, bar charts, and yearly trend charts.
SQL: validation tables for launch sites, payload mass, boosters, and outcomes.
Folium: launch-site maps, success/failure markers, and distance analysis.

How insights were used

EDA selected the most relevant explanatory variables for modeling.
Dash enabled interactive filtering by launch site and payload range.
Visual evidence supported the classification phase and final conclusions.

[GitHub sources: 4. jupyter-labs-eda-dataviz.ipynb · 5. jupyter-labs-eda-sql.ipynb · 6. lab_jupyter_launch_site_location.ipynb · spacex_dash_app.py](#)

From analysis to decision support

Learning curve signal

The yearly success trend suggests SpaceX improved landing reliability over time, so model validation should use newer launches whenever possible.

Operational segmentation

Launch site and orbit type should not be treated as background variables; they segment missions with different risk profiles.

Payload interaction

Payload mass alone does not fully explain outcome. Its relationship with success changes by orbit and launch site.

Decision-support use case

The model can support pre-launch risk screening and cost estimation, but it should complement engineering judgment rather than replace it.

These insights strengthen the project beyond the template by connecting EDA, interactive analytics, and classification to a practical business decision.

Rubric alignment before PDF submission

Required evidence

- Public GitHub URL is visible on slide 2.
- Notebook and Python file sources are listed by methodology.
- Final deliverable should be submitted as PDF.

Content completeness

- Methodology slides contain process explanations, not only titles.
- Result slides include visuals plus key analytical interpretation.
- Folium, Dash, SQL, EDA, and ML sections are all represented.

Additional value

- Discussion connects findings to decision support.
- Innovative insights highlight learning curve, segmentation, and payload interaction.
- Design is consistent and easy to follow.